WEEK 2 INTERNSHIP REPORT

Embedded Firmware Development and Simulation Testing

Project Title	ESP32-Based Smart Environmental Monitoring System with Temperature Alert Using DHT22
Student Name	_____
Internship / Organization	_____
Submission Date	_____

1. Objective

Develop and test firmware for a simulated embedded system, demonstrating GPIO control, periodic sensor acquisition, serial debugging, non-blocking timing using millis(), and a threshold-based alert.

2. Project Overview

An ESP32 and DHT22 sensor monitor temperature and relative humidity. Readings appear in the Serial Monitor. The LED turns ON at or above 30.0 °C and remains OFF below the threshold.

3. Hardware and Software Requirements

• ESP32 board simulated in Wokwi
• DHT22 temperature/humidity sensor
• LED and series current-limiting resistor
• Wokwi simulator and Serial Monitor
• Arduino-style C++ source in sketch.ino
• Adafruit DHT library and dependency if prompted

4. System Connections

DHT22 VCC → 3V3
DHT22 DATA/SDA → GPIO 4
DHT22 GND → GND
LED anode → GPIO 2 through a series resistor
LED cathode → GND

5. Firmware Design

setup() configures the LED, starts serial communication at 115200 baud, and initializes the DHT22. loop() uses millis() to schedule readings every 2,000 milliseconds without delay(). Invalid sensor readings trigger an error message and switch the LED OFF. Valid readings are printed; the LED is ON at or above 30.0 °C and OFF below it.

6. Algorithm

1. Initialize LED, serial communication, and sensor.
2. Check whether 2 seconds have elapsed.
3. Read temperature and humidity.
4. Handle invalid readings by displaying an error and turning the LED OFF.
5. Print valid readings.
6. Turn LED ON at/above 30.0 °C; otherwise turn it OFF.

7. Repeat continuously.

7. Simulation and Testing

Testing was performed in Wokwi. The user reported that both temperature-alert tests passed.

Test Results

Below 30 °C (e.g., 25 °C): expected LED OFF and normal-status message — **Passed**.
At/above 30 °C (e.g., 35 °C): expected LED ON and high-temperature alert — **Passed**.
Sensor reading schedule: approximately every 2 seconds — **Implemented**.

Attach screenshots of the circuit and Serial Monitor for both test conditions before submission.

8. Results and Learning Outcomes

• ESP32 GPIO output control
• DHT22 sensor interfacing in simulation
• Serial communication for readings and debugging
• Periodic non-blocking timing with millis()
• Threshold-based alert logic and basic error handling

9. Limitations and Future Improvements

This is a simulation, not physical hardware testing. The 30.0 °C threshold is a demonstration setting. Future improvements include an MQ-135 air-quality sensor, LDR, buzzer, Wi-Fi telemetry, and IoT dashboard.

10. Conclusion

The prototype demonstrates periodic temperature and humidity monitoring and an automatic LED alert. The reported Wokwi tests passed for normal and high-temperature conditions, providing a foundation for extending the Week 1 design.

Appendix A — Firmware Source

The complete commented firmware is provided separately as sketch.ino. README_Week_2.txt provides setup and simulation instructions.